

SYSTEM ANALYSIS AND DESIGN

Unit-III: System Requirement Specification and Analysis

Comprehensive Study Notes

Himachal Pradesh University · BCA Semester IV

UNIT-III: SYSTEM REQUIREMENT SPECIFICATION AND ANALYSIS

1. System Requirement Specification (SRS)

1.1 What is System Requirement Specification?

System Requirement Specification (SRS) is a document that describes the requirements of a proposed system in a clear, complete, and unambiguous manner.

Definition (Pressman): "SRS is a document that specifies the requirements of the system to be developed. It serves as a contract between the developer and the client."

1.2 Characteristics of Good SRS

Characteristic	Description
Correct	Accurately describes the system requirements
Complete	Covers all aspects of the system
Unambiguous	No ambiguity in interpretation
Consistent	No contradictions within the document
Verifiable	Requirements can be verified
Modifiable	Easy to change and update
Traceable	Requirements can be traced
Prioritized	Requirements are prioritized

1.3 Types of Requirements

+-----+	+-----+
FUNCTIONAL	NON-FUNCTIONAL
REQUIREMENTS	REQUIREMENTS
+-----+	+-----+
What the system	Quality attributes
should do	of the system
Business rules	Performance, security

| User interactions | | Reliability, usability |
+-----+ +-----+

1.4 Functional Requirements

Type	Description	Example
Business Requirements	High-level business needs	"System should generate sales reports"
User Requirements	User needs and expectations	"User should be able to enter order details"
System Requirements	Detailed system functions	"System shall validate the entered data"

1.5 Non-Functional Requirements

Type	Description
Performance	Response time, throughput, capacity
Security	Authentication, authorization, data protection
Reliability	Availability, fault tolerance, recovery
Usability	Ease of use, user interface
Maintainability	Ease of maintenance and updates
Portability	Ability to run on different platforms

2. Fact Finding Techniques

2.1 What are Fact Finding Techniques?

Fact Finding Techniques are methods used to gather information about the current system and user requirements for a new system.

+-----+	+-----+
INTERVIEWS	QUESTIONNAIRES
+-----+	+-----+
+-----+	+-----+
OBSERVATION	DOCUMENT ANALYSIS
+-----+	+-----+

2.2 Interviews

Interviewing is one of the most effective fact-finding techniques for gathering information directly from stakeholders.

Type	Description
Structured Interview	Pre-determined questions with specific answers
Unstructured Interview	Open-ended questions, flexible format
Semi-structured Interview	Combination of both approaches
Group Interview	Interviewing multiple people together

Advantages of Interviews

Advantage	Description
Rich Information	Detailed and in-depth information
Flexibility	Can adapt based on responses

Clarification	Can clarify ambiguous answers
Relationship Building	Builds rapport with stakeholders
Understanding	Better understanding of issues

Disadvantages of Interviews

Disadvantage	Description
Time-Consuming	Requires significant time
Costly	Expensive in terms of time and resources
Subjectivity	May be influenced by personal biases
Scheduling	Difficult to schedule with busy people

2.3 Questionnaires

Questionnaires are written sets of questions distributed to users to gather information.

Advantages

Advantage	Description
Reach	Can reach a large number of people
Cost-Effective	Less expensive than interviews
Time-Saving	Saves time for both parties
Consistency	Same questions for all respondents
Anonymity	Respondents can remain anonymous

Disadvantages

Disadvantage	Description
Low Response Rate	Many people don't respond
Limited Depth	Cannot probe for details
Misinterpretation	Questions may be misunderstood
Incomplete Answers	Questions may be left unanswered

2.4 Observation

Observation involves watching users perform their tasks to understand the current system.

Advantages

Advantage	Description
Real Data	Gathers actual behavior
Objective	Less subjective than interviews
Reveals Issues	Identifies hidden problems
Work Flow	Understands work flow

Disadvantages

Disadvantage	Description
Time-Consuming	Requires significant time
Hawthorne Effect	People change behavior when observed
Limited Scope	Only captures what is observed
Interruption	May interrupt work flow

2.5 Document Analysis

Document Analysis involves examining existing documents and reports.

Document Type	Examples
Organizational Documents	Organization chart, policies, procedures
Business Documents	Invoices, reports, forms

System Documents	Existing system documentation
External Documents	Industry standards, regulations

2.6 Fact Finding Comparison

Technique	Strengths	Weaknesses
Interviews	Rich information, flexible	Time-consuming, costly
Questionnaires	Cost-effective, wide reach	Low response rate, limited depth
Observation	Objective, real data	Time-consuming, Hawthorne effect
Document Analysis	Objective, historical data	May be outdated

3. Data Flow Diagram (DFD)

3.1 What is a Data Flow Diagram?

A **Data Flow Diagram (DFD)** is a graphical representation of the flow of data through a system. It shows how data moves from input to output.

Definition: "DFD is a graphical technique that depicts the flow of data through a system and the processes that transform the data."

3.2 DFD Components

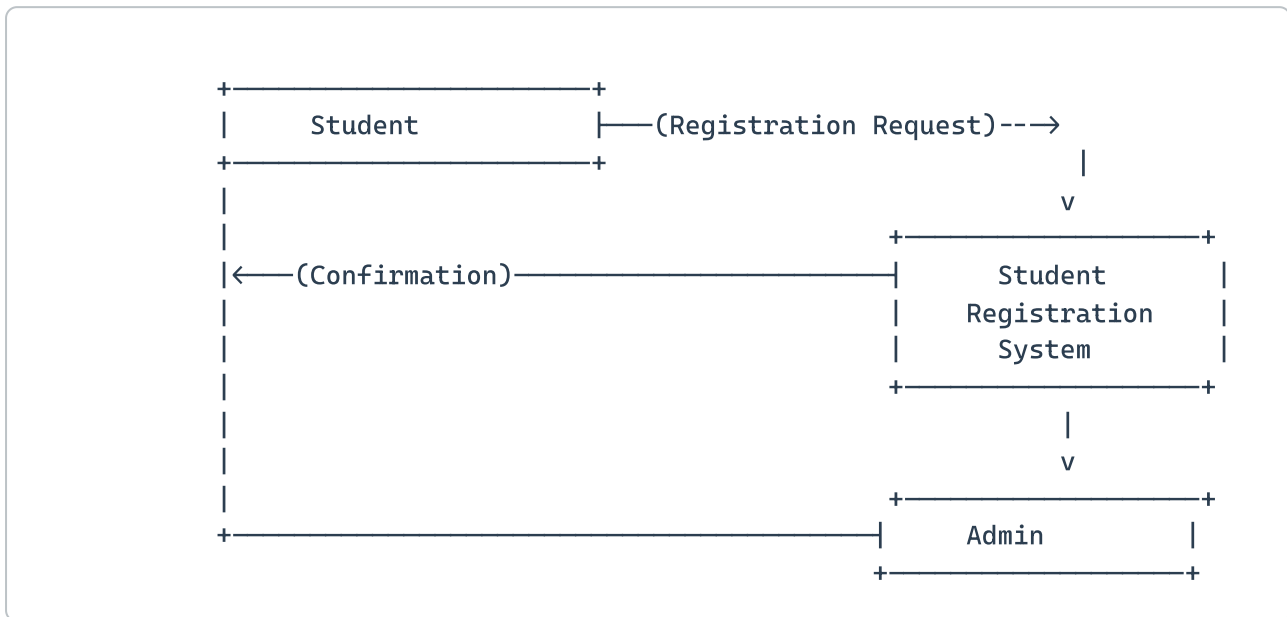
Component	Symbol	Description
Process	Circle / Rounded Rectangle	Transforms input to output
External Entity	Rectangle	Source or destination of data
Data Flow	Arrow	Movement of data
Data Store	Open Rectangle	Storage of data

3.3 Levels of DFD

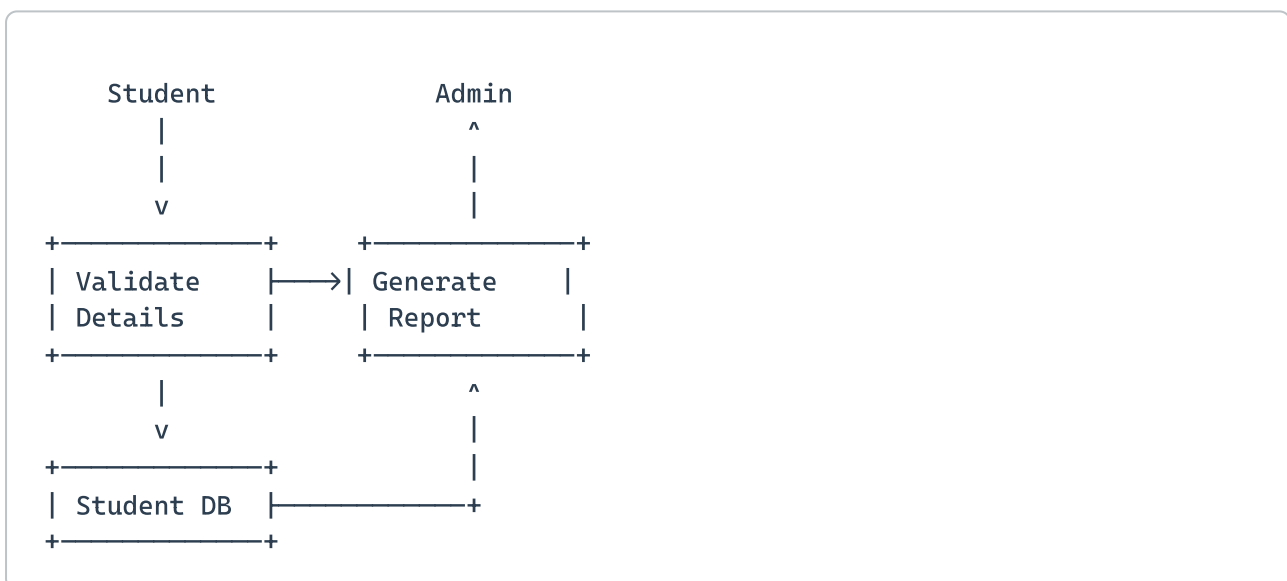
Level	Name	Description
Level 0	Context Diagram	Shows entire system as a single process with external entities and data flows
Level 1	Level 1 DFD	Breaks the context diagram into major processes showing internal data flows and data stores
Level 2	Level 2 DFD	Further decomposes level 1 processes into more detailed sub-processes

3.4 Solved Example: Student Registration System

Context Diagram (Level 0)



Level 1 DFD



4. Data Dictionary

4.1 What is a Data Dictionary?

A **Data Dictionary** is a centralized repository of information about data elements used in the system. It describes the data items, their structure, and relationships.

Definition: *"Data dictionary is a repository that contains metadata about the data elements in a system."*

4.2 Data Dictionary Components

Component	Description
Data Element Name	Name of the data item
Description	Detailed description of the data
Data Type	Type of data (text, number, date)
Length	Maximum length of data
Format	Specific format requirements
Source	Where the data comes from
Valid Values	Allowed values
Uses	Where the data is used

4.3 Data Dictionary Example

Data Element	Description	Type	Length	Format
Student_ID	Unique student identifier	Text	10	XXX9999999

Student_Name	Full name of student	Text	50	Alphabetic
Date_of_Birth	Student date of birth	Date	10	DD/MM/YYYY
Course_Code	Course identifier code	Text	8	XXX-999
Marks	Marks obtained in course	Number	3	0-100

4.4 Elements of Data Dictionary

- **Data Flow:** Description of data movement between processes, stores, and entities
- **Data Store:** Description of where data is stored within the system
- **Data Structure:** Logical arrangement and relationships among data elements
- **Process Specification:** Detailed description of how data is transformed by processes

5. Decision Analysis

5.1 What is Decision Analysis?

Decision Analysis is the process of evaluating and choosing among alternatives. It is used when decisions need to be made based on multiple criteria.

DECISION TREES	DECISION TABLES
Graphical	Tabular
Hierarchical	Matrix
Shows decisions and consequences	Shows conditions and actions

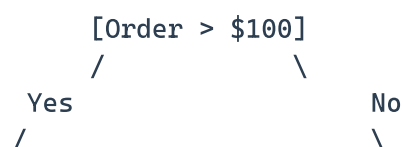
5.2 Decision Trees

Decision Trees are graphical representations of decision-making processes. They use a tree-like structure where each internal node represents a condition, each branch represents a decision outcome, and each leaf represents an action.

Construction of Decision Tree

1. Identify the decision to be made
2. Identify all possible conditions or criteria
3. Arrange conditions hierarchically from root to leaves
4. Draw branches for each possible outcome of a condition
5. Assign actions at leaf nodes representing final decisions

Decision Tree Example





5.3 Decision Tables

Decision Tables are matrix representations that show the relationship between conditions and actions. They tabulate all possible combinations of conditions and the corresponding actions to be taken.

Types of Decision Tables

- **Limited Entry Decision Table:** Conditions are expressed as Boolean (Yes/No) values; actions are either taken or not taken
- **Extended Entry Decision Table:** Conditions can have multiple values beyond Yes/No; actions can have multiple outcomes

Construction Steps for Decision Table

1. Identify all conditions (the decision criteria)
2. Identify all possible actions (the decisions)
3. Determine the number of rules (2^n for n conditions in limited entry)
4. Fill condition entries for all combinations
5. Specify action entries for each rule
6. Reduce the table by combining redundant rules

Decision Table Example

Conditions / Rules	R1	R2	R3	R4
Order > \$100?	Y	Y	N	N
Loyal Customer?	Y	N	Y	N
Actions				

10% Discount	X	-	-	-
5% Discount	-	X	X	-
No Discount	-	-	-	X

5.4 Decision Tree vs Decision Table

Aspect	Decision Tree	Decision Table
Representation	Graphical tree structure	Tabular matrix format
Ease of Understanding	Easy to visualize and follow	Compact and systematic
Number of Conditions	Suitable for fewer conditions	Suitable for many conditions
Modification	Difficult to modify	Easy to modify
Redundancy Check	Not inherent	Easy to check redundancies

6. Structured English

6.1 What is Structured English?

Structured English is a method of describing program logic and processing requirements using a subset of English language combined with structured programming constructs. It bridges the gap between natural language and programming code.

6.2 Structured English Constructs

IF-THEN-ELSE Structure

```
IF customer is loyal THEN
    Apply 10% discount
ELSE
    Apply 5% discount
ENDIF
```

DO-WHILE Structure

```
DO WHILE there are unprocessed orders
    Calculate order total
    Apply discount if applicable
    Generate invoice
ENDDO
```

CASE Structure

```
CASE OF customer_type
    CASE "Premium":
        Apply 20% discount
    CASE "Gold":
        Apply 15% discount
    CASE "Silver":
        Apply 10% discount
    CASE "Regular":
        Apply 5% discount
    CASE ELSE:
```




No discount
ENDCASE

6.3 Advantages of Structured English

- Easy to understand by non-technical stakeholders
- Language-independent and implementation-agnostic
- Provides a clear, logical representation of business rules
- Can be easily converted into programming code
- Useful for documenting complex decision logic

7. Design Approaches

7.1 Top-Down Approach

Top-Down Approach starts at the highest level and breaks down into smaller components. The system is first designed at a high level of abstraction and then decomposed into more detailed sub-systems and modules.

Characteristics

- Starts from the system level and proceeds downward
- High-level modules are designed first
- Details are added incrementally
- Emphasizes overall system understanding

7.2 Bottom-Up Approach

Bottom-Up Approach starts at the lowest level and combines components into larger systems. Individual components are developed first and then integrated to form higher-level systems.

Characteristics

- Starts from the component level and proceeds upward
- Low-level modules are developed first
- Components are integrated to build larger units
- Emphasizes reuse of existing components

7.3 Comparison: Top-Down vs Bottom-Up

Aspect	Top-Down	Bottom-Up
Start	From the top (system level)	From the bottom (component level)
Direction	High to low	Low to high

Focus	Overall view	Specific components
Advantage	Good for planning	Good for integration
Disadvantage	May miss details	May miss overall view

8. Audit Trail

8.1 What is an Audit Trail?

An **Audit Trail** is a chronological record of system activities that allows the reconstruction of events. It is used for security, compliance, and problem investigation.

8.2 Types of Audit Trails

Type	Description
System Audit Trail	Records system-level activities
Application Audit Trail	Records application-level activities
User Audit Trail	Records user activities
Data Audit Trail	Records data changes
Security Audit Trail	Records security events

8.3 Importance of Audit Trails

Importance	Description
Security	Detects unauthorized access
Compliance	Meets regulatory requirements
Problem Investigation	Helps investigate issues
Accountability	Holds users accountable
Data Integrity	Ensures data integrity

9. Questions for Practice

Short Answer Questions (2-5 Marks)

1. What is system requirement specification?
2. What are the types of requirements?
3. What are fact finding techniques?
4. What is a data flow diagram?
5. What is a data dictionary?
6. What is a decision tree?
7. What is a decision table?
8. What is the top-down approach?
9. What is the bottom-up approach?
10. What is an audit trail?

Long Answer Questions (10 Marks)

1. Explain system requirement specification and its characteristics.
2. Explain the different fact finding techniques.
3. Explain data flow diagrams with examples.
4. Explain data dictionaries with examples.
5. Explain decision trees and decision tables.
6. Explain top-down and bottom-up approaches.

Objective Questions (1 Mark)

1. SRS stands for:
(a) System Requirement Specification (b) Software Requirement Specification (c) Both
(d) None
2. DFD stands for:
(a) Data Flow Diagram (b) Data Flow Design (c) Data Format Diagram (d) None

3. Interviews are a:
(a) Fact finding technique (b) Design technique (c) Testing technique (d) None
4. Top-down approach starts from:
(a) Top level (b) Bottom level (c) Middle level (d) None
5. Audit trail is used for:
(a) Security (b) Compliance (c) Both (d) None

End of Unit-III Notes · SYSTEM ANALYSIS AND DESIGN · BCA Sem IV (HPU)